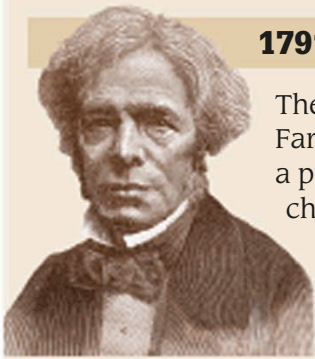




1791–1867 MICHAEL FARADAY

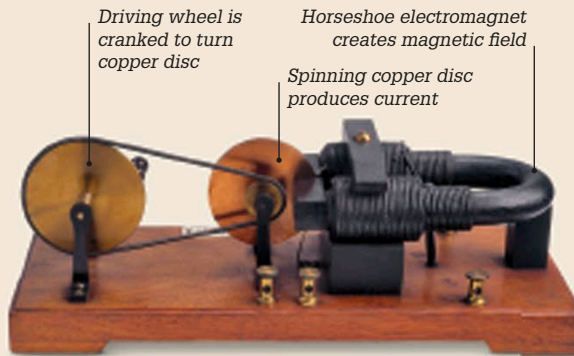
The son of a London blacksmith, Michael Faraday taught himself science before securing a post as assistant to Humphry Davy, an English chemist. Faraday became the greatest scientist of his day. His discovery of electromagnetic induction led to electricity being generated and used in many everyday applications.

Man of science

As well as investigating electromagnetism, Faraday discovered benzene, a chemical compound, and established the laws of electrolysis (the chemical reactions that occur when an electric current passes through a liquid).

Generating electricity

Faraday was able to produce a weak electric current by spinning a copper disc within a magnetic field (electromagnetic induction). In time, his discovery led to the development of machines that could generate large quantities of electricity.



Model of Faraday's disc generator



Replica of Faraday's electromagnetic induction ring

Coils of copper wire carrying electric current wrapped in cotton

Driving wheel is cranked to turn copper disc

Horseshoe electromagnet creates magnetic field

Spinning copper disc produces current

KEY DATES

1825 Faraday was appointed director of the laboratory at the Royal Institution.

1831 He discovered electromagnetic induction.

1833 Faraday published his laws of electrolysis.

1830

Gradual change

In his book *Principles of Geology*, Scottish geologist Charles Lyell argued that Earth was more than 300 million years old and had gone through many geological ages. This theory of the Earth's history contradicted Cuvier's belief in catastrophes (see p.122).



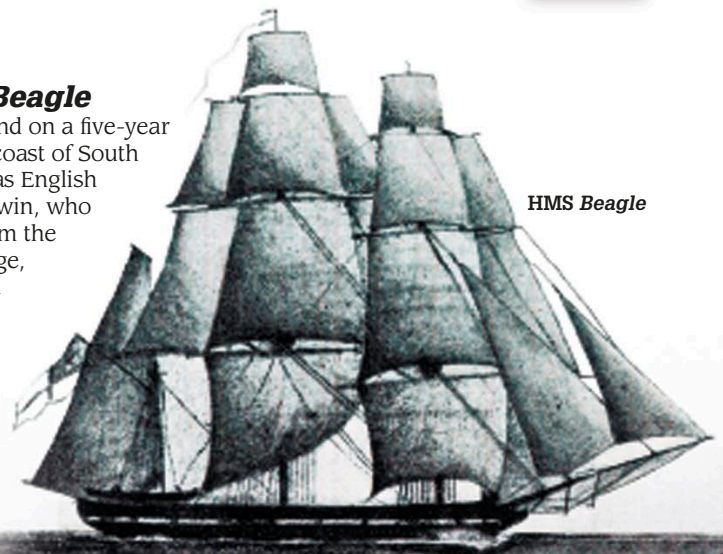
Charles Lyell

1831

Voyage of the *Beagle*

HMS *Beagle* left England on a five-year voyage to survey the coast of South America. On board was English naturalist Charles Darwin, who had just graduated from the University of Cambridge, England. He took with him a copy of Lyell's *Principles of Geology*.

Charles Darwin
See pages 134–135



HMS *Beagle*



William Whewell, an English professor, invented the word “scientist” in 1833.

1835